

REPORT

*Logic Gate Layout, Layers, GDS, and
Fabrication*

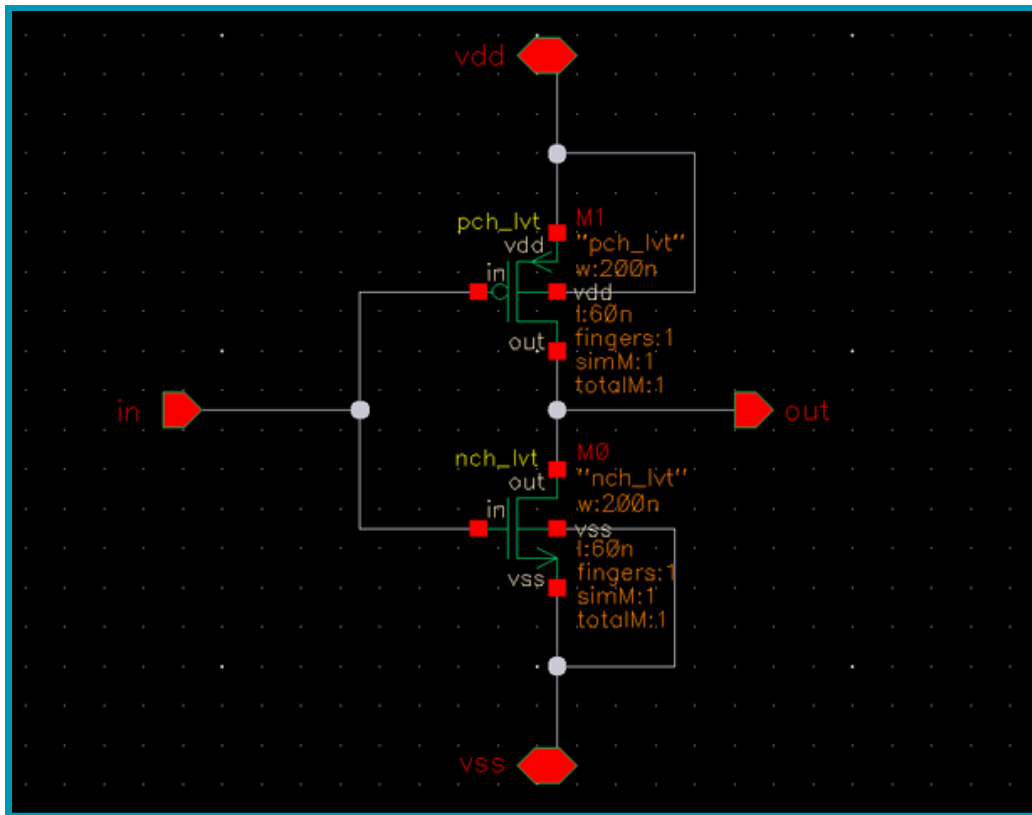


Submitted by

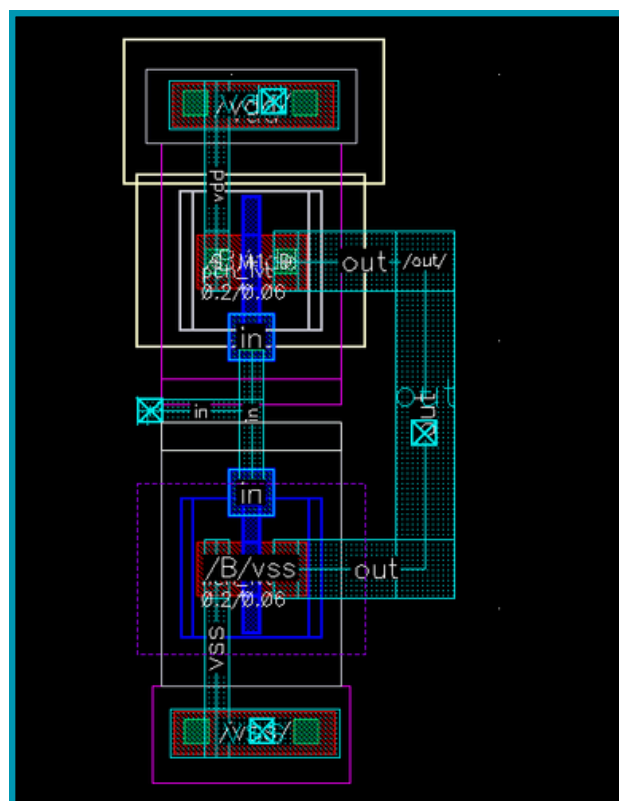
Mai Sherif Fekry

Task1

a. The schematic view showing the sizing

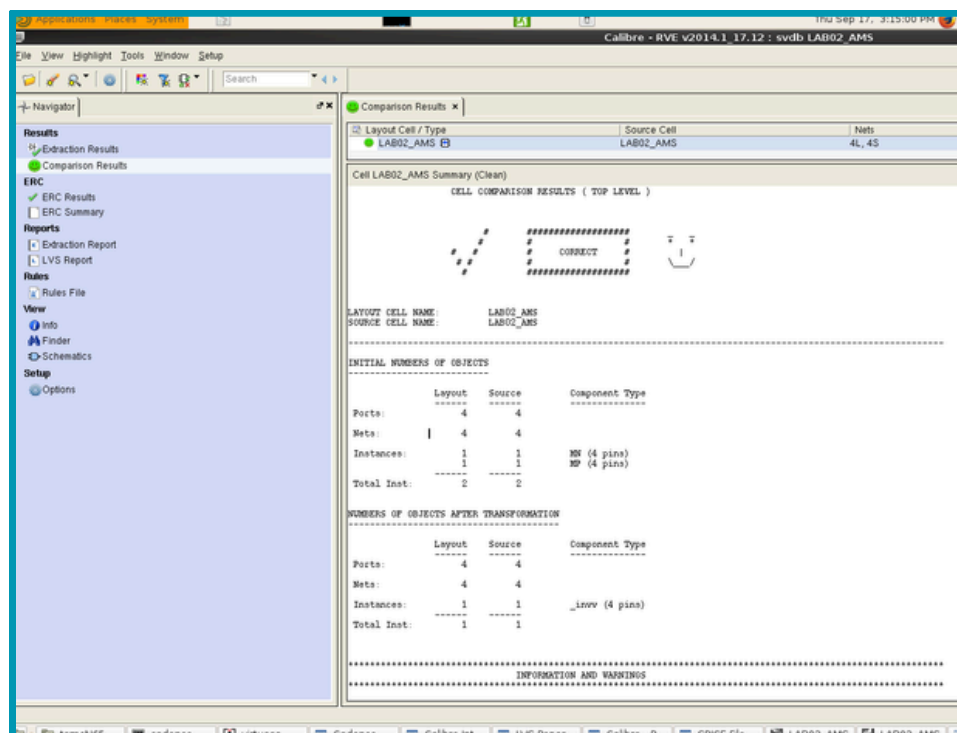
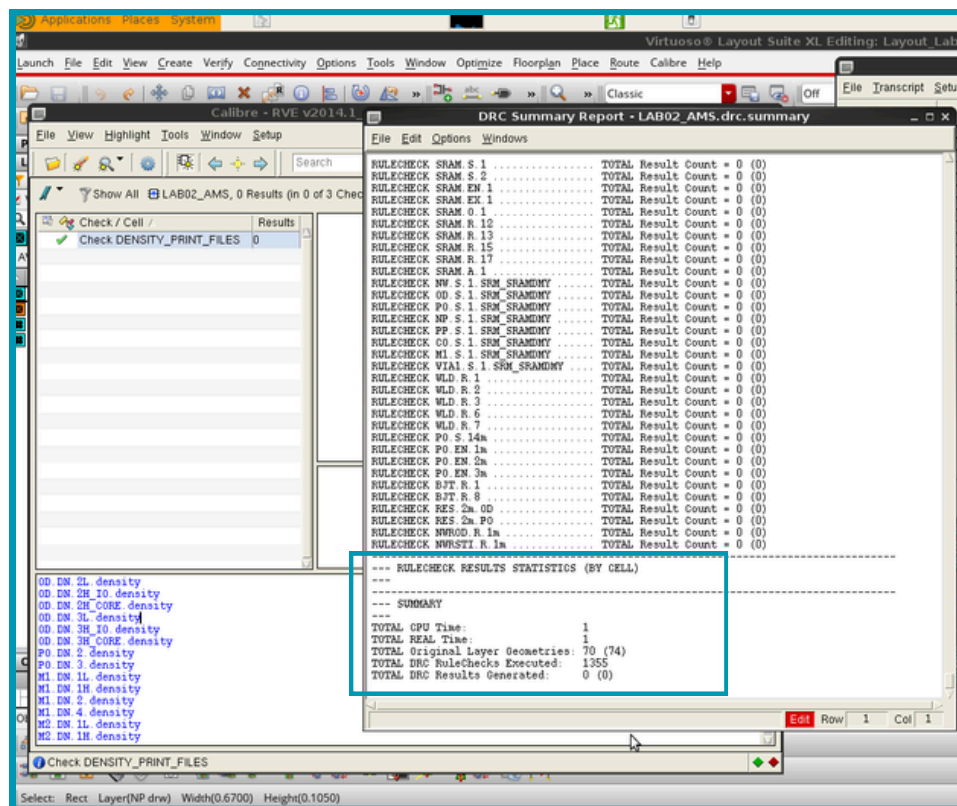


b. The layout after passing LVS and after the DRC final state.



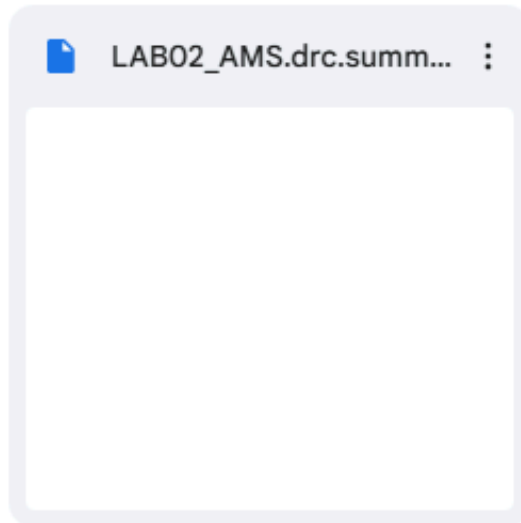
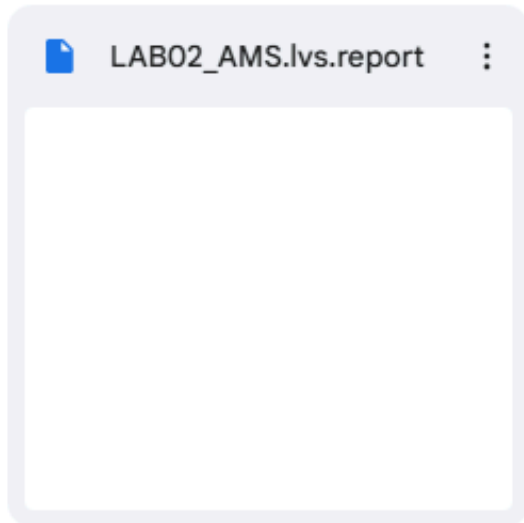
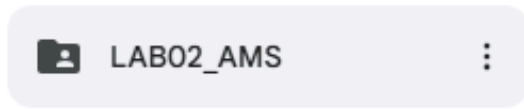
Task1

c. The final DRC and LVS final reports.

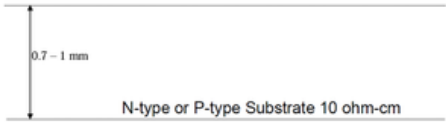
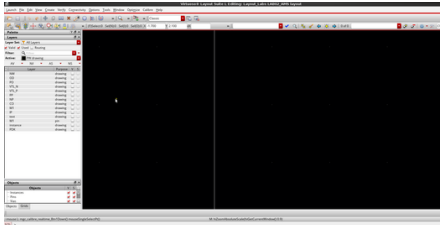
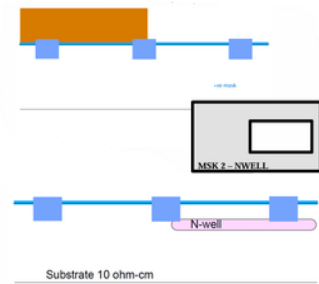
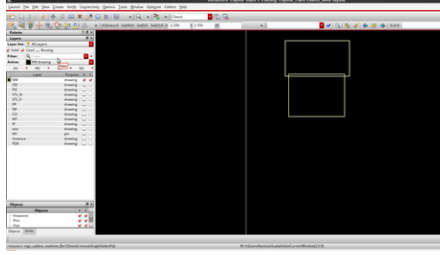
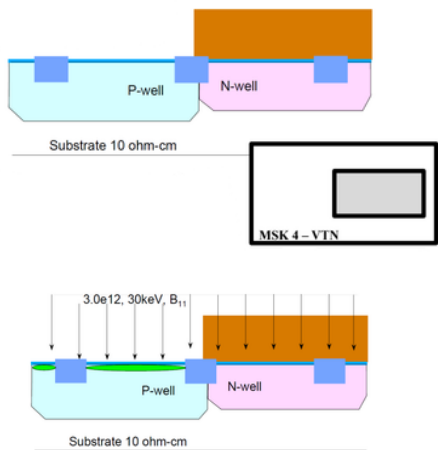
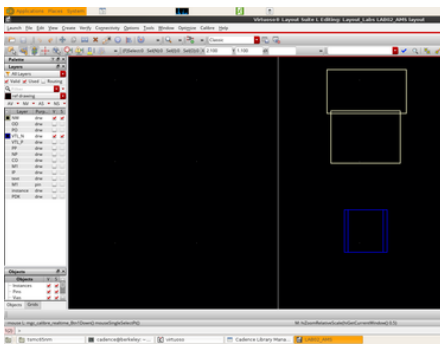


Task1

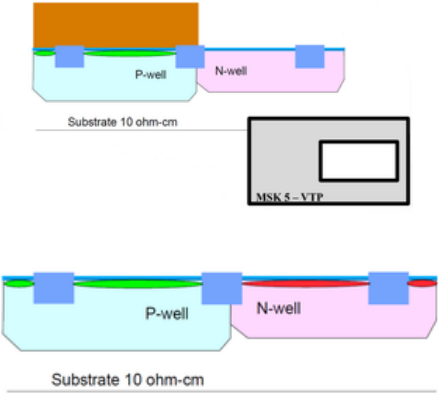
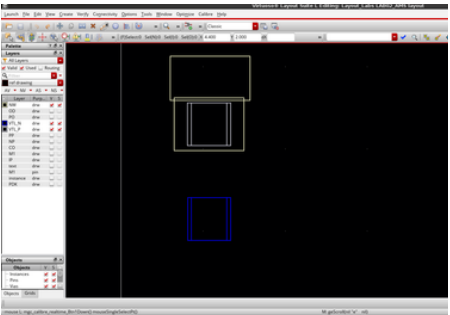
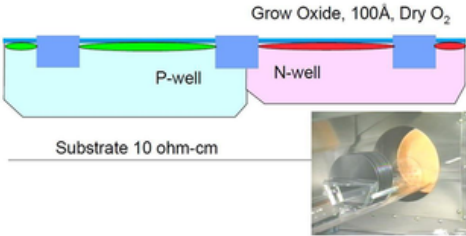
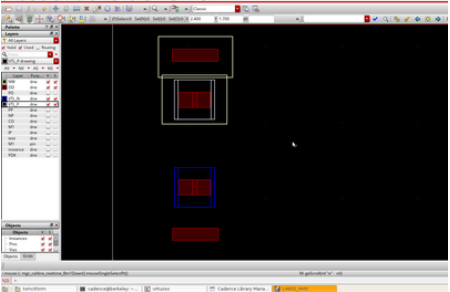
d. Attached the LVS and DRC final reports



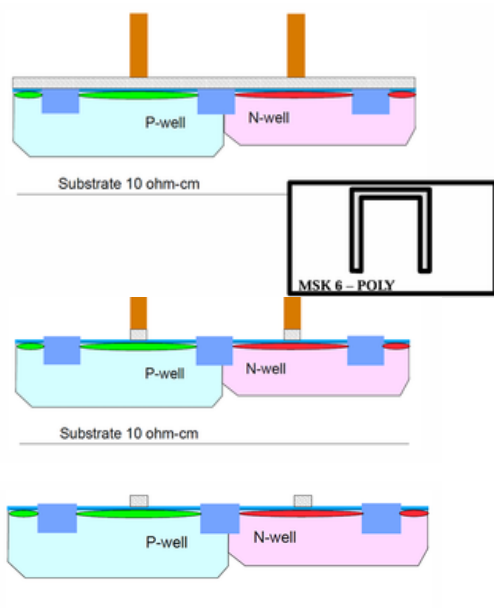
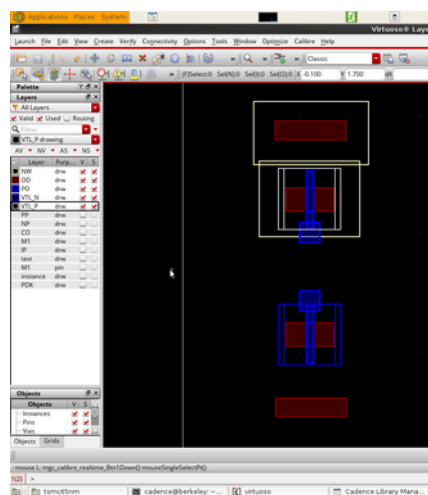
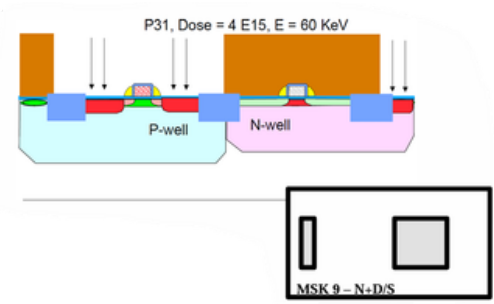
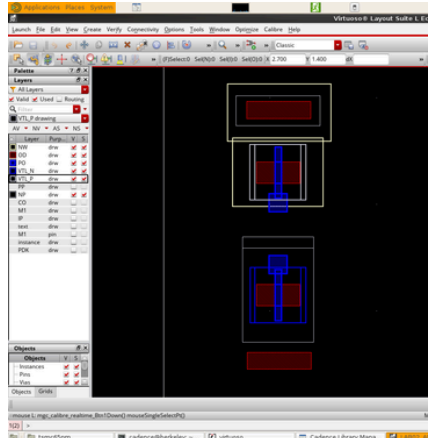
Task2

Fabrication Process Stages	Layer	Mask Used (TSMC)
 Start with blank wafer	 Pwell	-
 Expose photoresist through n-well mask	 NWELL	NW1V
 Implantation needed to get desired threshold voltage.	 NMOS VT Adjust Implant	VTL_N

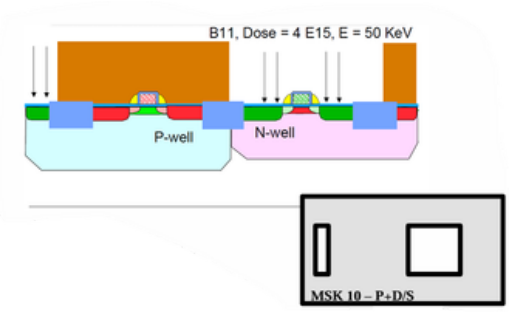
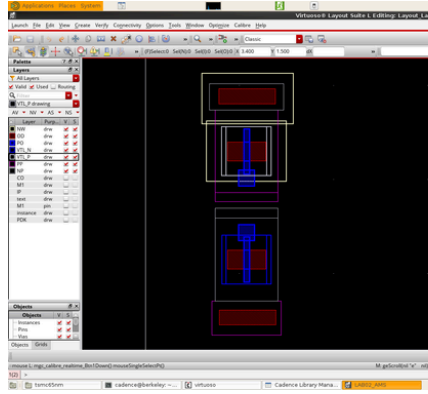
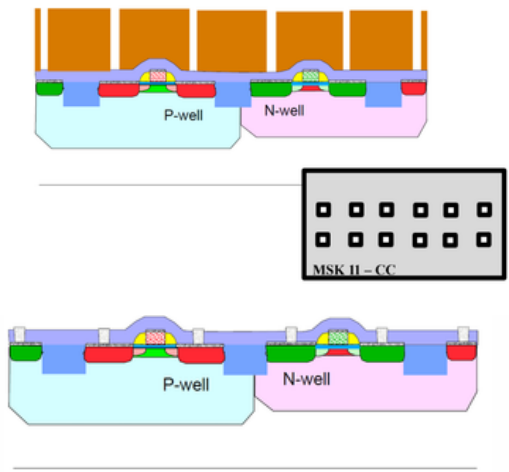
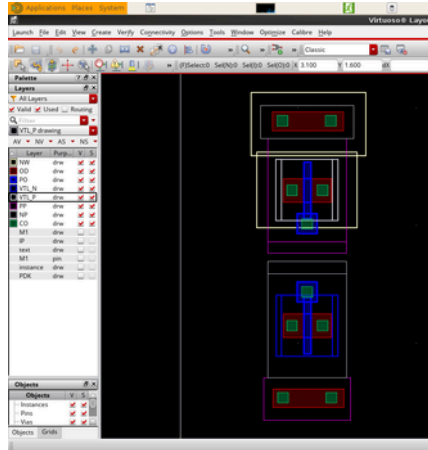
Task2

Fabrication Process Stages	Layer	Mask Used (TSMC)
 Substrate 10 ohm-cm P-well N-well MSK 4 - VTP PMOS VT Adjust	 Low Threshold Voltage Layer	VTL_P
 Grow Oxide, 100Å, Dry O₂ Substrate 10 ohm-cm P-well N-well Gate Oxide Growth	 Oxide Diffusion, p- and n-type active areas	OD

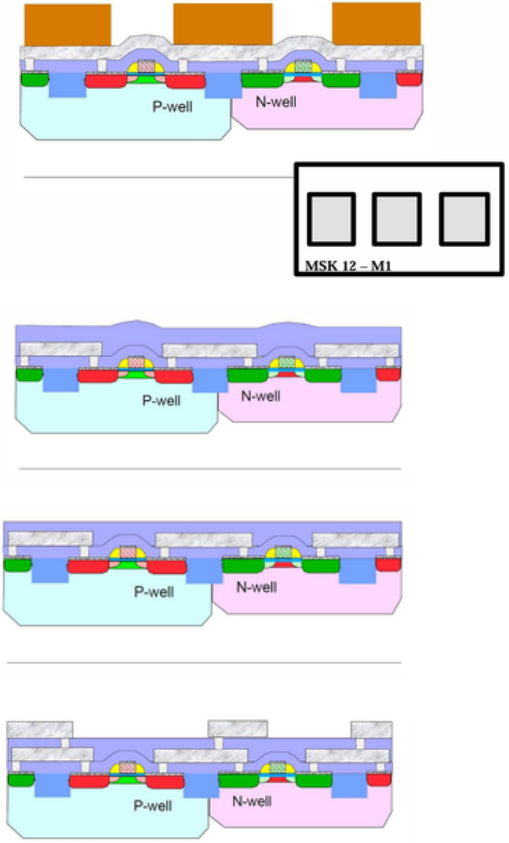
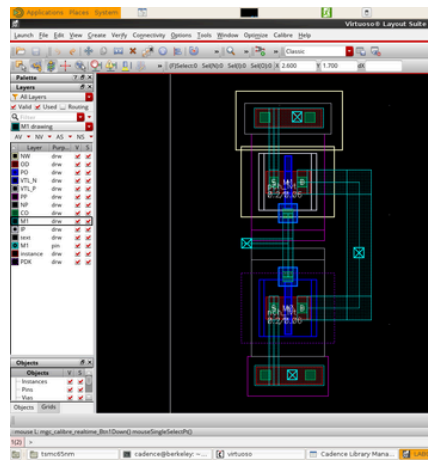
Task2

Fabrication Process Stages	Layer	Mask Used (TSMC)
 Poly-Silicon Deposition and Etching and PR Stripping	 Polysilicon Layer	PO
 N+D/S Ion Implantation	 N+ drain and source for NMOS and PMOS bulk diffusion	NP

Task2

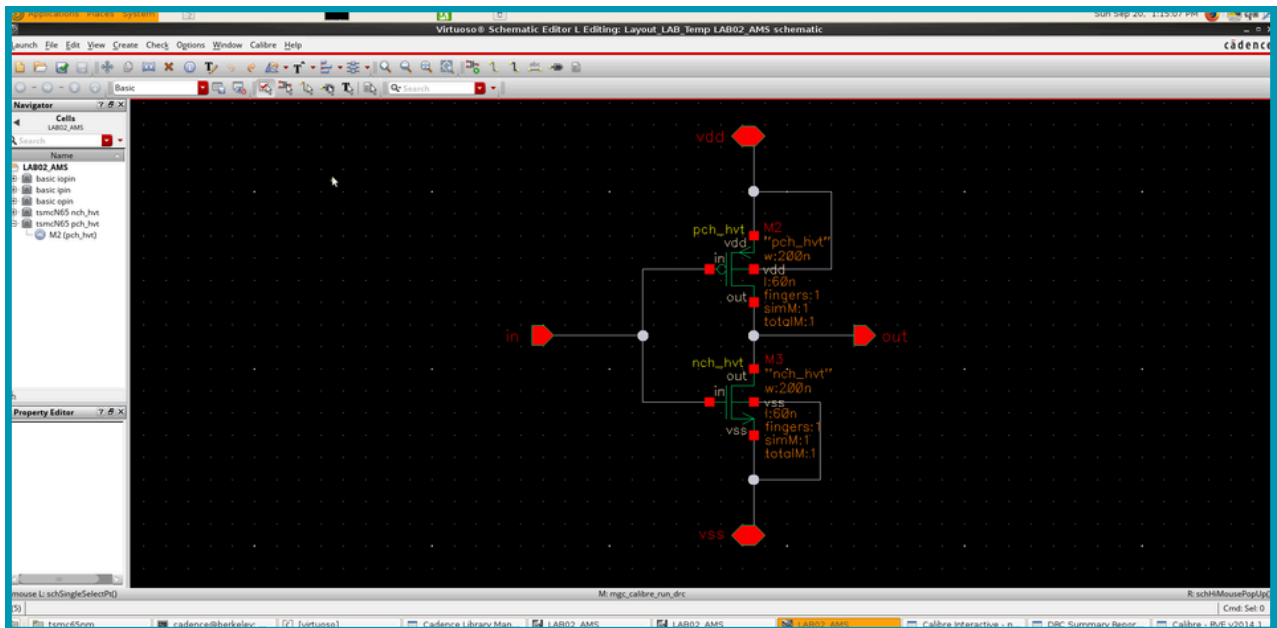
Fabrication Process Stages	Layer	Mask Used (TSMC)
 P+D/S Ion Implantation	 N+ drain and source for PMOS and NMOS bulk diffusion	PP
 Contact cut mask	 CO	CO

Task2

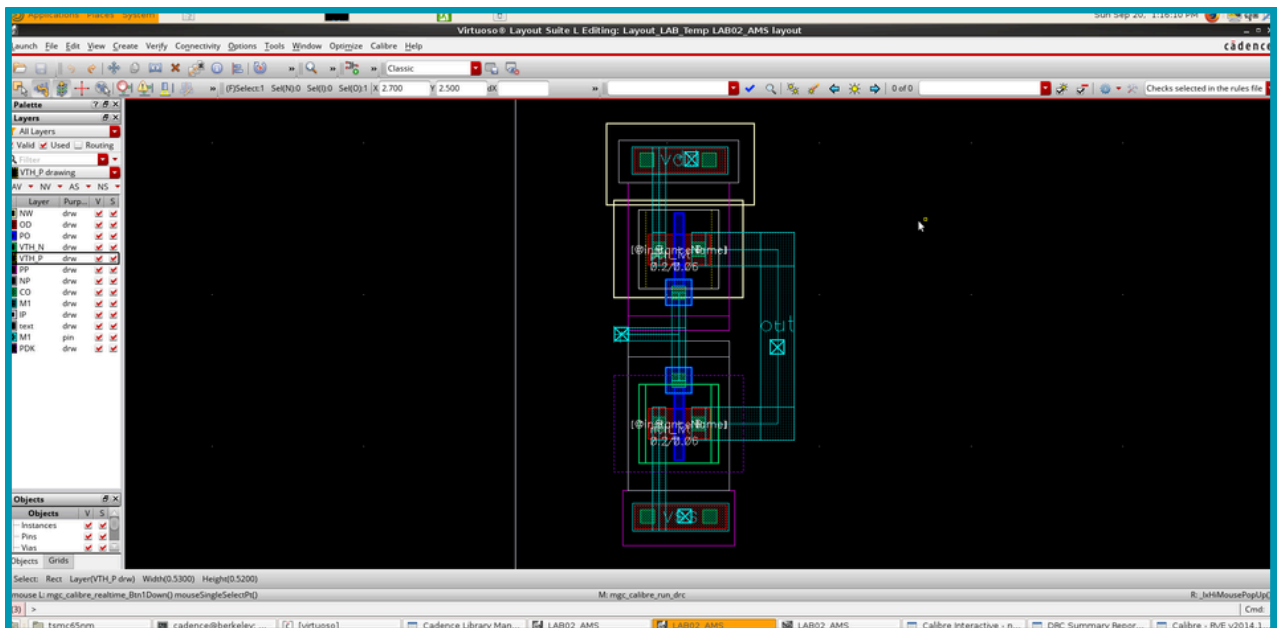
Fabrication Process Stages	Layer	Mask Used (TSMC)
 MSK 12 - M1 Aluminium deposition and etching	 M1 Layer	M1

Task3

a. The schematic view



b. The layout view



Task3

c. The modified layer map file highlighting the modification

I changed the names of the VTL_N, P to VTH_N, P such that these numbers in the stream (.gds) file to be mapped to those layers instead of the low threshold volatge layers

17	OD	test3	6	13
18	PDIF	drawing	7	0
19	NDIFF	drawing	8	0
20	NT_N	drawing	11	0
21	NT_N	Ncap_NTN	11	20
22	VTL_N	drawing	12	0
23	VTL_P	drawing	13	0
24	OD_33	drawing	15	0
25	OD_33	drawing	15	0

Before Modification

17	OD	test3	6	13
18	PDIF	drawing	7	0
19	NDIFF	drawing	8	0
20	NT_N	drawing	11	0
21	NT_N	Ncap_NTN	11	20
22	VTH_N	drawing	12	0
23	VTH_P	drawing	13	0
24	OD_33	drawing	15	0

After Modification

d. Attach the LVS, DRC final reports and the modified layer map file

